Natural logarithm

1 Find the value of each of the following correct to four decimal places:

a $\ln 94$ b $\ln 0.042$ c $\ln 78^4$ d $\ln(18 \times 35)$

2 Consider $x = \ln 31$. Find the value of x, correct to two decimal places.

3 Use the properties of logarithms to express each of the following without any powers or surds:

a $\ln y^2$ b $\ln \frac{1}{y^2}$ c $\ln \sqrt{y}$ d $\ln \sqrt[3]{y}$

4 Use the properties of logarithms to rewrite $-2\ln x$ with a positive power.

5 Rewrite each of the following as the sum and difference of logarithms:

a $\ln\left(\frac{15}{8}\right)$ b $\ln\left(\frac{pq}{r}\right)$ c $\ln\left(\frac{19}{7}\right)$

6 Rewrite each of the following expressions as a single logarithm:

a $\ln 3 + \ln 5$ b $\ln 24 - \ln 4$ c $\ln 8 - \ln 32$ d $\ln(3x) + \ln(5y)$

7 Which expression is equivalent to $2\ln(7x)$ for $x > 0$?

A $\ln(14x)$ **B** $\ln 14 + \ln x$ **C** $\ln 49 + \ln x$ **D** $\ln(49x^2)$

8 Simplify:

a $e^{\ln w}$ b $\ln(e^x)$ c $2^{\ln e^y}$ d $e^{\ln x} \times e^{\ln y}$

e $\ln\left(\frac{e^x}{e}\right)$ f $\ln e^x - \ln e^y$ g $\ln\left(\frac{1}{e^x}\right)$ h $\ln\left(\frac{e^5}{e^x}\right)$

9 Evaluate each of the following expressions:

a $\ln e^{3.5}$ b $\ln e^4$ c $\sqrt{6}\ln\left(e^{\sqrt{6}}\right)$ d $\ln\left(\frac{1}{e^2}\right)$